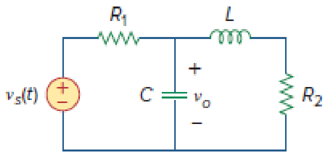
Problem

Using Fig. 16.56, design a problem to help other students understand how to use Thevenin's theorem (in the *s*-domain) to aid in circuit analysis.

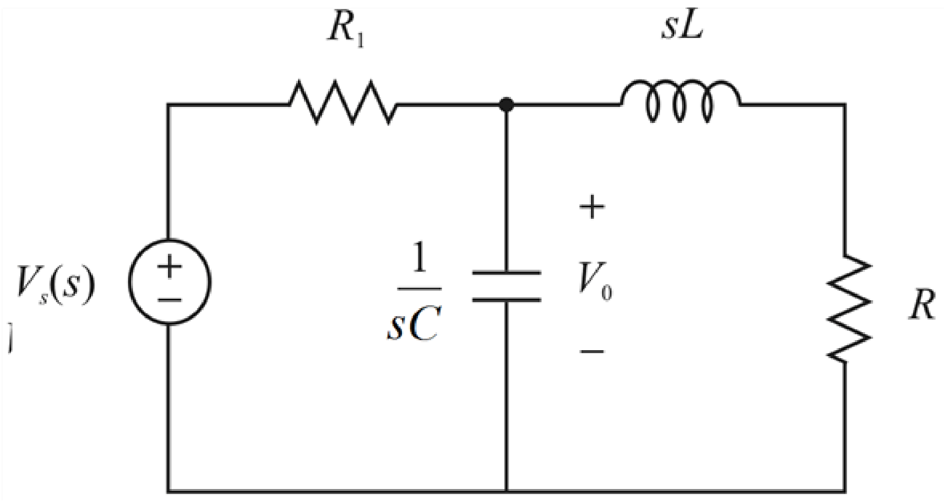
Figure 16.56:



Step-by-step solution

Step 1 of 7

The circuit in *s*-domain



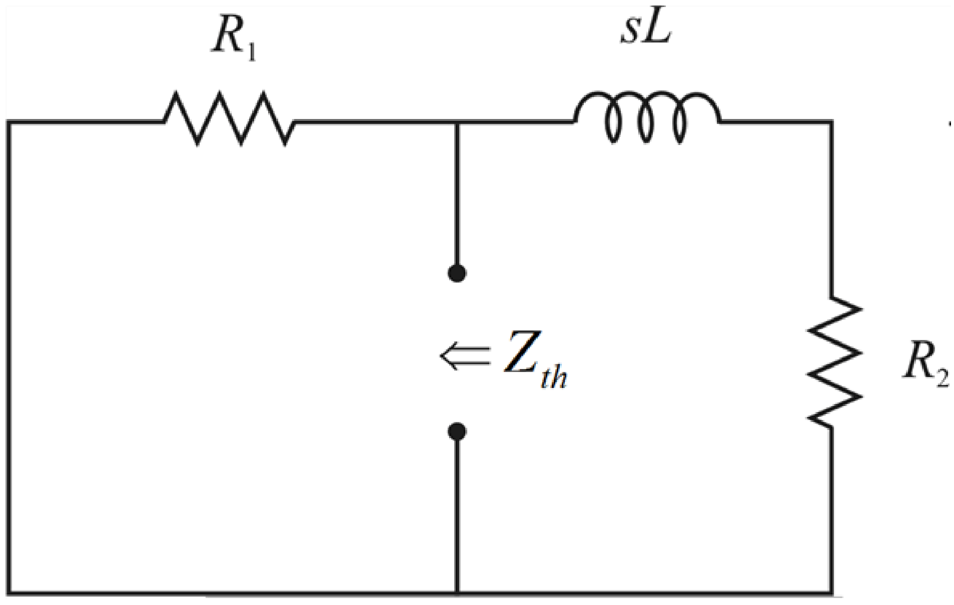
Step 2 of 7

$i_L(0)=0$ zero for all $t<0$

$v_C(0)=0$ for $t<0$

Step 3 of 7

For the Thevenin impedance



Step 4 of 7

sL inductor and R_2 resistor are in series and in parallel with R_1 resistor

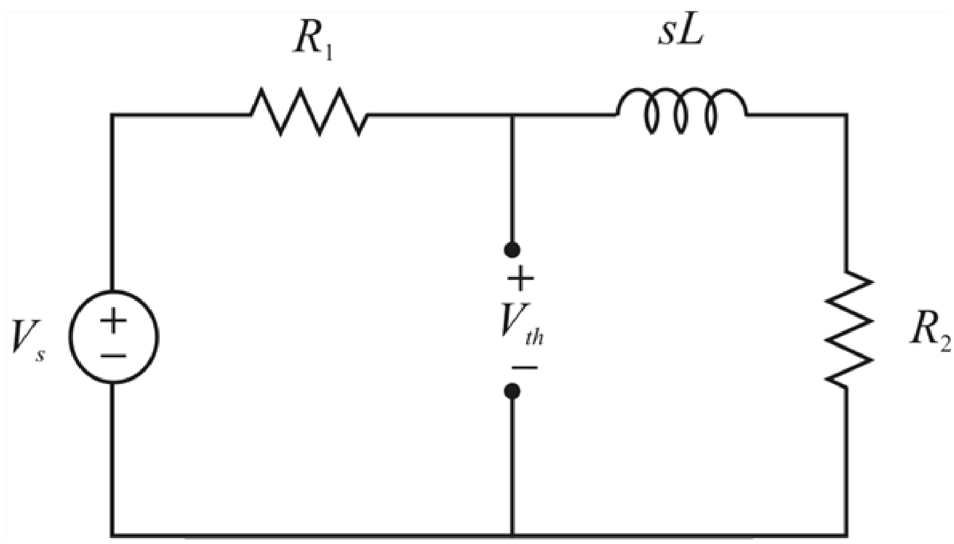
$$Z_{th} = R_1 \| (sL + R_2)$$

$$Z_{th} = \frac{R_1(sL + R_2)}{R_1 + R_2 + sL}$$

$$Z_{th} = \frac{R_1\left(s + \frac{R_2}{L}\right)}{s + \left(\frac{R_1 + R_2}{L}\right)}$$

Step 5 of 7

Thus, the Thevenin voltage by the circuit



Step 6 of 7

Using nodal analysis, we get

$$\frac{V_{th}-V_s}{R_1}+\frac{V_{th}}{sL+R_2}=0$$

$$V_{th}\left(\frac{1}{R_1}+\frac{1}{sL+R_2}\right)=\frac{V_s}{R_1}$$

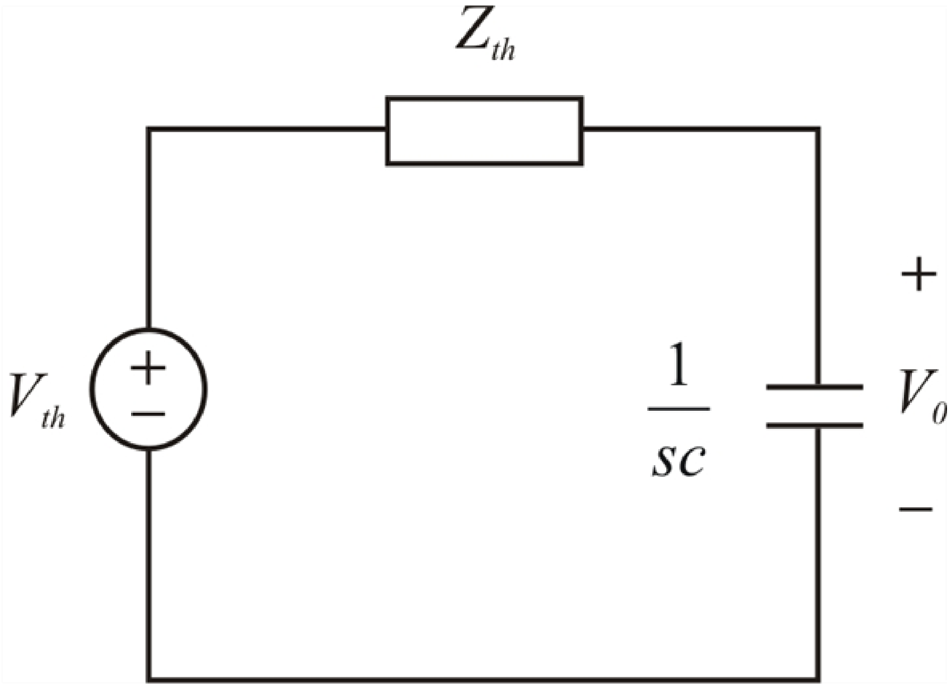
$$V_{th}\left(\frac{sL+R_2+R_1}{R_1\left(sL+R_2\right)}\right)=\frac{V_s}{R_1}$$

$$V_{th}=\frac{V_s\left(sL+R_2\right)}{sL+R_2+R_1}$$

$$V_{th}=\left[\frac{s+\frac{R_2}{L}}{s+\frac{R_1+R_2}{L}}\right]V_s$$

Step 7 of 7

Thus, the Thevenin circuit is



By the voltage division method,

$$V_0 = \frac{\frac{1}{sC}}{\frac{1}{sC} + Z_{th}} V_{th}$$

$$V_0 = \left[\frac{\frac{1}{sC}}{\frac{1}{sC} + \frac{R_1\left(s + \frac{R_2}{L}\right)}{s + \frac{R_1 + R_2}{L}}} \right] \left[\frac{s + \frac{R_2}{L}}{s + \frac{R_1 + R_2}{L}} \right]$$

$$V_0 = \left[\frac{s + \frac{R_1 + R_2}{L}}{s + \frac{R_1 + R_2}{L} + s^2 R_1 C + s\frac{R_2 R_1 C}{L}} \right] \left[\frac{s + \frac{R_2}{L}}{s + \frac{R_1 + R_2}{L}} \right]$$

$$V_0 = \frac{s + \frac{R_2}{L}}{s^2 R_1 C + s\left(1 + \frac{R_1 R_2 C}{L}\right) + \left(\frac{R_1 + R_2}{L}\right)}$$

Assume the values and substitute in the expression and solve using partial fraction expansion, then by taking inverse transform we get the required output voltage $v_0(t)$.